

IBM BOB AI INNOVATION HACKATHON 2026

# GridHealth AI

Predictive grid-health monitoring and maintenance prioritisation for power utilities

TEAM TELEMETRY TITANS · TRACK: AI/ML

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## THE PROBLEM

# Utilities still react to blackouts instead of preventing them

Transformer and substation failures still cause blackouts costing utilities \$1M+ per hour and affecting millions of people — while the data needed to prevent them already exists.

**\$1M+**

lost per hour of  
blackout downtime

**Weeks**

of advance warning hidden  
in sensor data

**0**

sensor + weather feeds  
combined today

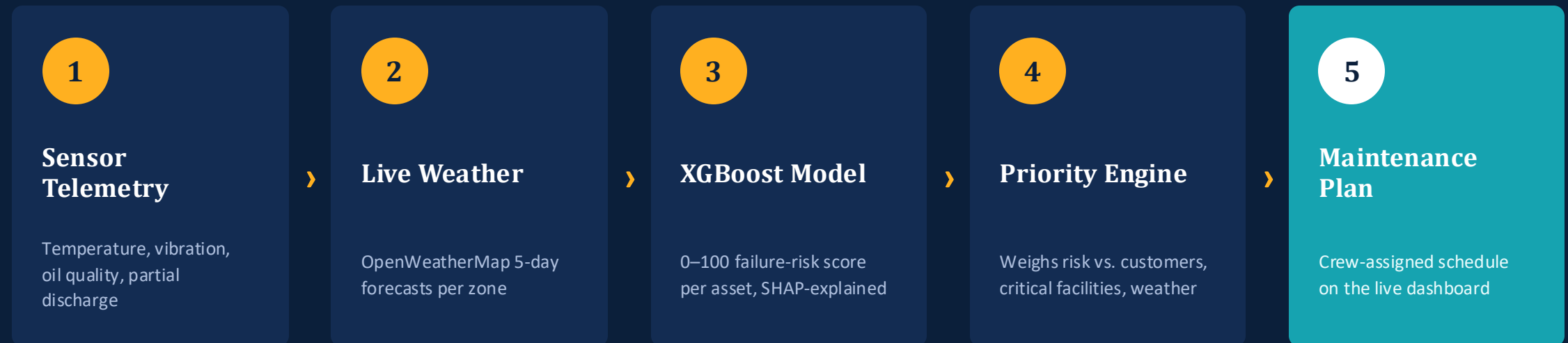
## Root causes

- Calendar-based maintenance ignores real asset condition
- Temperature, vibration, oil quality & partial-discharge signals are collected but rarely analysed together
- Weather forecasts and grid risk are managed in separate silos, so compounding risk goes unseen

## THE SOLUTION

# One pipeline, from raw telemetry to a crew-assigned plan

GridHealth AI fuses sensor telemetry, live weather forecasts and three years of incident history into a single risk-ranked view of every grid asset — explained, prioritised and scheduled automatically.



## KEY FEATURES

# Six capabilities that turn data into action

1

### XGBoost Failure Prediction

Trained on 30-day sensor trends + incident history; scores 0–100 failure risk per asset.

2

### Priority-Weighted Ranking

Composite score blends model risk, customers served, critical-facility and weather bonuses.

3

### SHAP Explainability

Per-asset feature-contribution breakdown, so engineers know why an asset is flagged.

4

### Live Weather Integration

Real OpenWeatherMap forecasts classified into typed alerts and overlaid on risk scores.

5

### Interactive Grid Map

Leaflet map with colour-coded risk markers across 5 geographic zones.

6

### Auto-Generated Maintenance

Risk-tiered scheduling (Day 0 → Day 13) with round-robin crew assignment.

## HOW PRIORITY IS CALCULATED

# A single composite score decides what gets fixed first



### Then the schedule writes itself

**CRITICAL**

Day 0

**HIGH**

Days 1–2

**MEDIUM**

Days 3–6

**LOW**

Days 7–13

# A full-stack, single-database platform

## Backend

FastAPI 0.111 · Uvicorn  
SQLAlchemy 2.0 · Pydantic v2

## ML / AI

XGBoost 2.0 · SHAP 0.45  
scikit-learn · pandas · NumPy

## Frontend

React 18 · Vite · TypeScript 5  
Tailwind CSS · React Router v7

## Frontend Libraries

Leaflet · Recharts  
TanStack Table · Zustand · Axios

## Data & Integrations

SQLite (grid\_data.db)  
OpenWeatherMap API

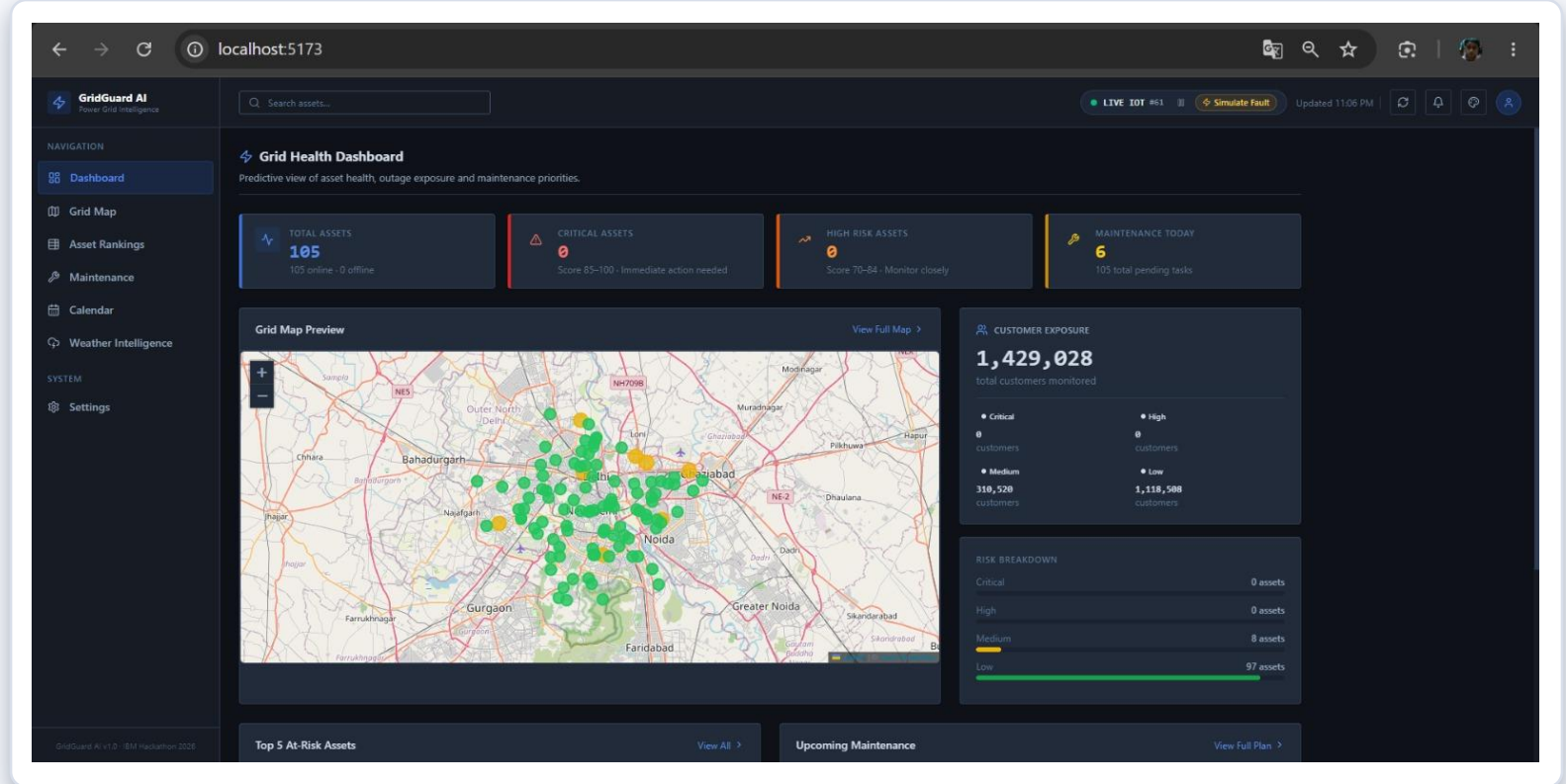
## Built With

IBM Bob — AI coding assistant  
used throughout development

*Single SQLite database shared by data pipeline, backend and frontend — runnable from scratch in under 5 minutes with three commands.*

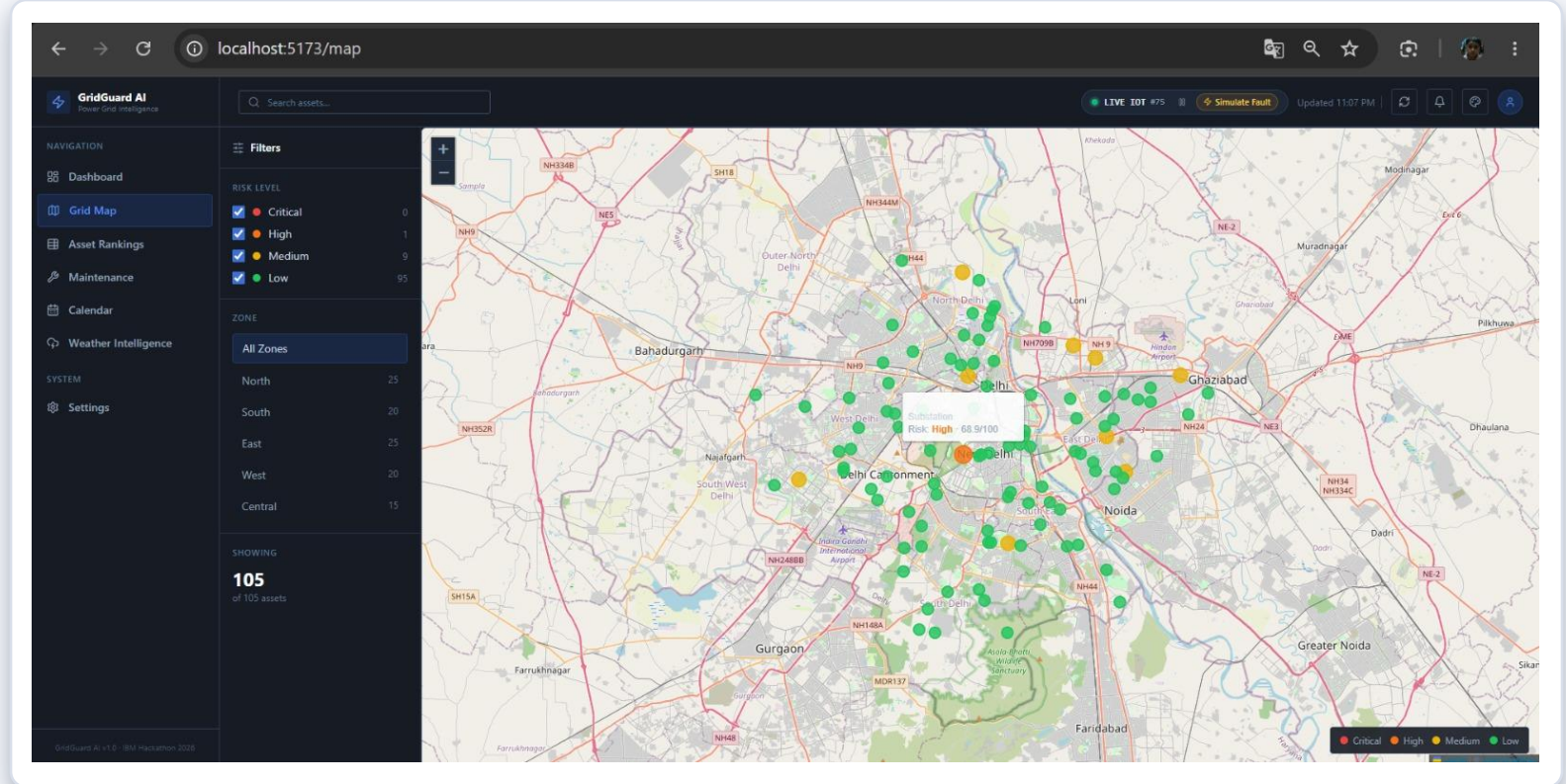
# Grid Health Dashboard

- At-a-glance view of all 105 monitored assets and their online status
- Live risk breakdown by Critical / High / Medium / Low tier
- 1.4M+ customers tracked by exposure across the network
- Grid map preview with one click through to the full map



# Interactive Grid Map

- Every asset plotted with a colour-coded risk marker
- Filterable by risk level and by one of 5 geographic zones
- Click-through detail popups show live risk score per asset
- Built on Leaflet / react-leaflet over OpenStreetMap tiles



# Asset Risk Rankings

- Sortable, filterable table across all 105 assets
- Risk score, status, customers served and 7-day failure % side by side
- Filter by risk level, zone, status or asset type
- One-click CSV export for offline reporting

GridGuard AI  
Power and intelligence

localhost:5173/assets

LIVE 10T #91 Simulate Fault Updated 11:07 PM

### Asset Risk Rankings

Prioritized view of grid equipment risk - 105 assets monitored

Export CSV

Filters Search by ID, name, zone... All Risk Levels All Zones All Statuses All Types Reset

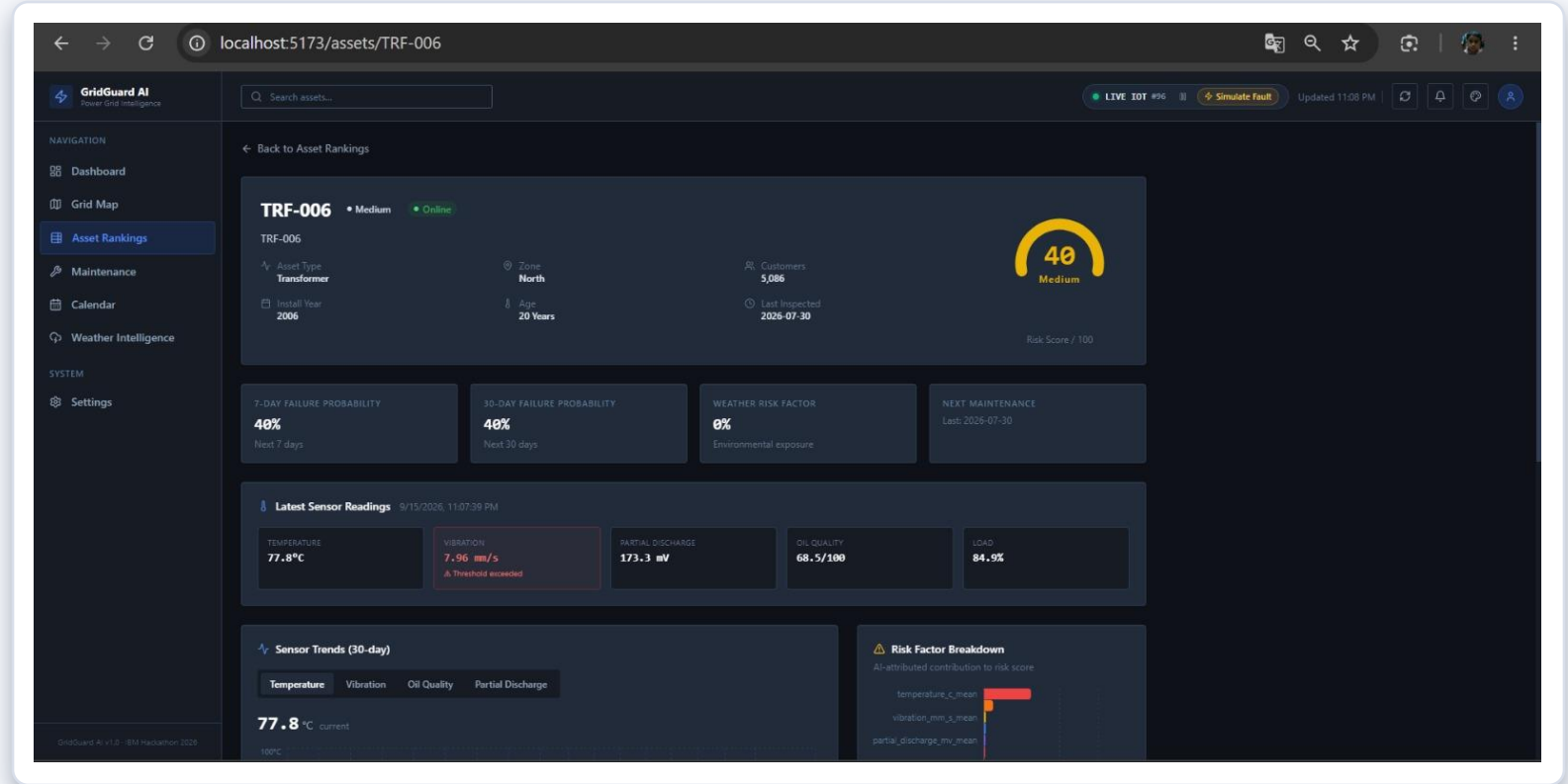
Search by ID, name, zone... 105 assets

| #  | ASSET ID | TYPE        | ZONE    | RISK SCORE | RISK LEVEL | STATUS | CUSTOMERS | LAST INSPECTED | 7D FAILURE | ACTION    |
|----|----------|-------------|---------|------------|------------|--------|-----------|----------------|------------|-----------|
| 01 | TRF-006  | Transformer | North   | 40         | Medium     | Online | 5,886     | 2026-07-30     | 40%        | Details → |
| 02 | TRF-033  | Transformer | East    | 38.8       | Medium     | Online | 6,369     | 2026-07-19     | 39%        | Details → |
| 03 | SUB-008  | Substation  | North   | 12         | Medium     | Online | 46,228    | 2026-03-24     | 12%        | Details → |
| 04 | SUB-033  | Substation  | Central | 12         | Medium     | Online | 45,364    | 2026-06-04     | 12%        | Details → |
| 05 | SUB-016  | Substation  | East    | 12         | Medium     | Online | 41,899    | 2026-09-04     | 12%        | Details → |
| 06 | SUB-019  | Substation  | East    | 12         | Medium     | Online | 38,057    | 2026-01-29     | 12%        | Details → |
| 07 | SUB-018  | Substation  | East    | 12         | Medium     | Online | 36,381    | 2026-07-13     | 12%        | Details → |
| 08 | SUB-007  | Substation  | North   | 12         | Medium     | Online | 34,969    | 2026-04-09     | 12%        | Details → |
| 09 | SUB-002  | Substation  | North   | 12         | Medium     | Online | 34,243    | 2026-08-11     | 12%        | Details → |
| 10 | SUB-027  | Substation  | West    | 12         | Medium     | Online | 33,387    | 2026-03-20     | 12%        | Details → |
| 11 | SUB-004  | Substation  | North   | 12         | Low        | Online | 31,591    | 2026-03-24     | 12%        | Details → |

GridGuard AI v1.0 - BM Hackathon 2026

# Asset Detail & SHAP Explainability

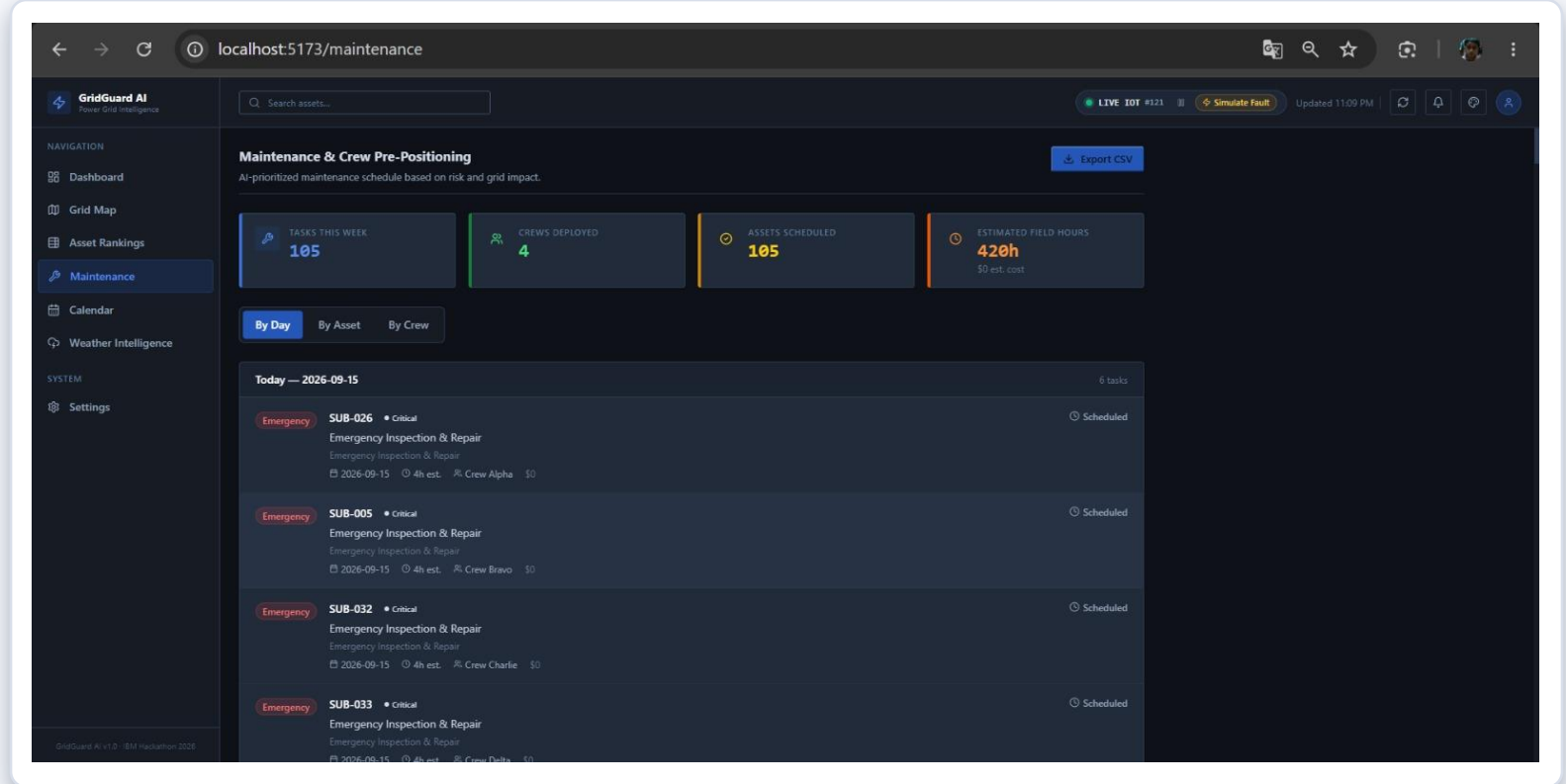
- Full asset profile: age, zone, customers, install year, last inspection
- 7-day and 30-day failure probability with live risk gauge
- Latest sensor readings with threshold-exceeded alerts
- 30-day sensor trend charts plus a SHAP feature-contribution breakdown



## PRODUCT WALKTHROUGH

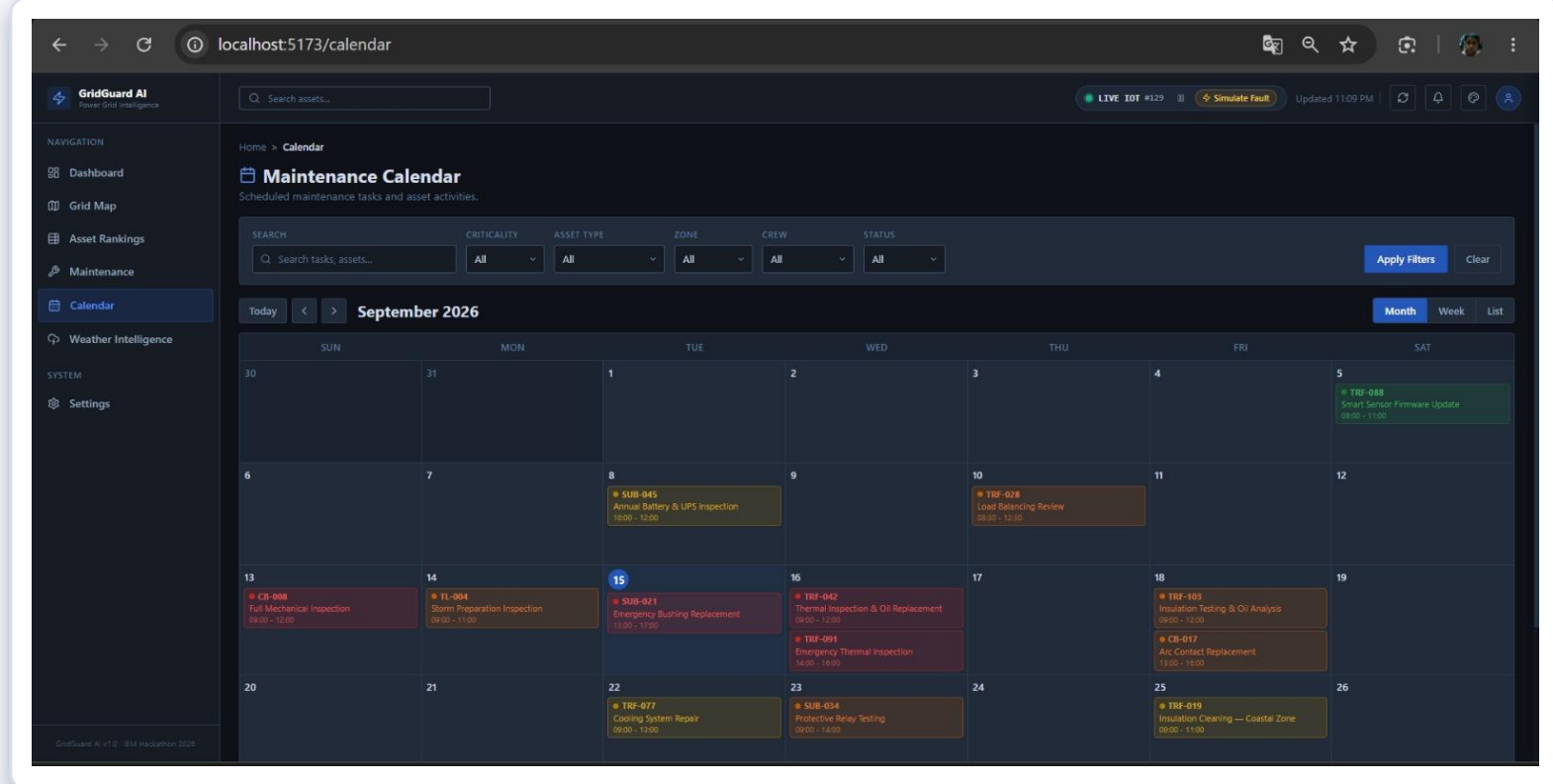
# Maintenance & Crew Pre-Positioning

- AI-prioritised schedule generated automatically from risk scores
- Round-robin assignment across crews, grouped by day
- Live counters for tasks, crews deployed and estimated field hours
- Switchable By Day / By Asset / By Crew views, exportable to CSV



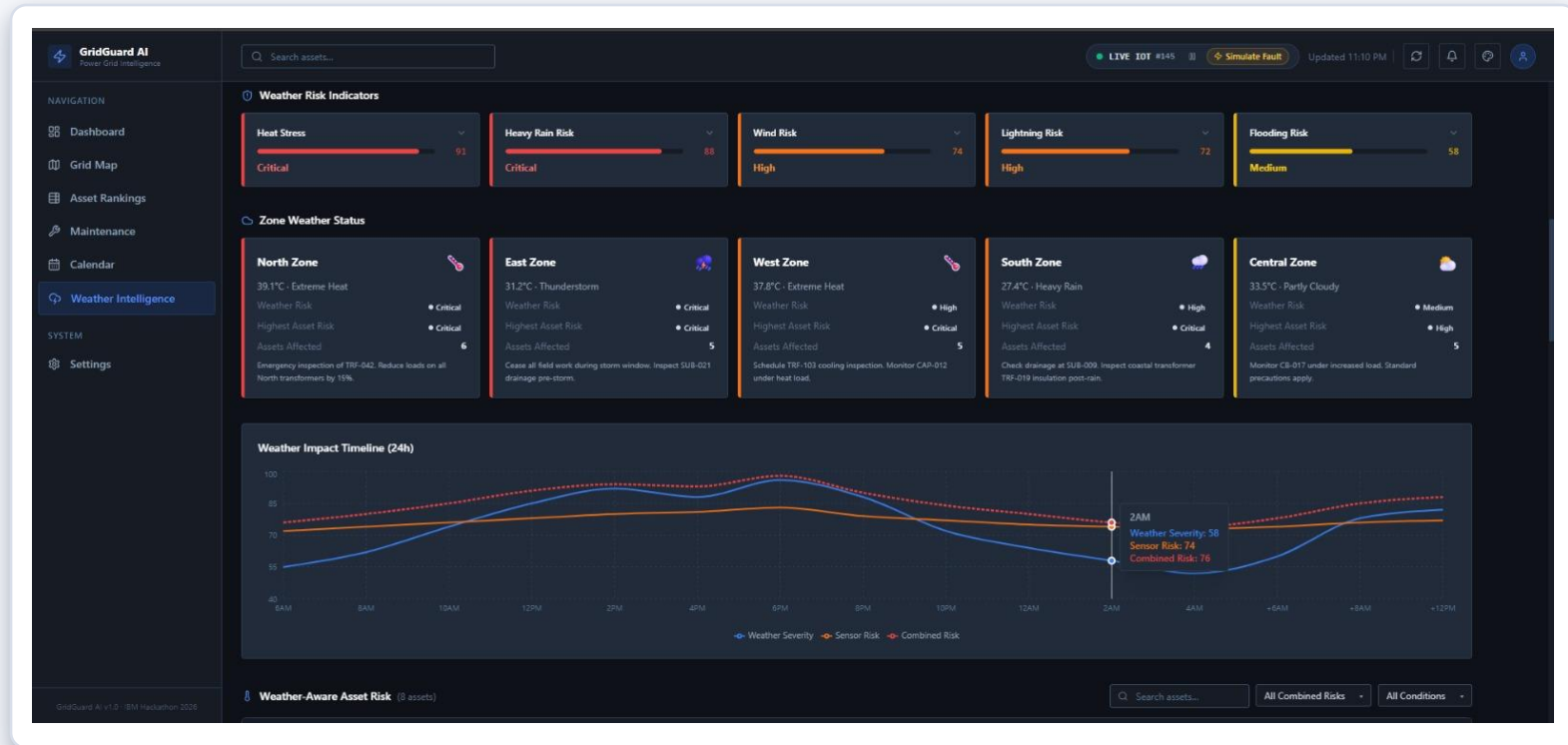
# Maintenance Calendar

- Month, week and list views of every scheduled task
- Colour-coded by urgency, from routine inspection to emergency repair
- Filterable by criticality, asset type, zone, crew and status
- One place to see today's emergencies alongside next month's routine work



# Weather Intelligence

- Live conditions and 5-day forecast per zone from OpenWeatherMap
- Heat, rain, wind, lightning and flood risk indicators, scored live
- Combined-critical counter shows assets re-evaluated under weather stress
- Per-zone recommended actions, e.g. reduce load, inspect drainage



# What's real today, and what's next

## Known Limitations

- Model trained on synthetic, rule-based labels — no real SCADA outage ground truth yet
- Live weather endpoint needs a free OpenWeatherMap key; falls back to DB-backed weather without one
- No authentication/authorisation — not production-ready
- Asset Detail sensor-trend charts currently use mock data, not the live sensors endpoint
- Tested on Chrome & Firefox; mobile layout works but isn't fully optimised

## What We're Proud Of

A complete, integrated pipeline: synthetic sensor data built on real transformer-failure physics, feeding an XGBoost model, explained with SHAP, served through FastAPI, and visualised in React — all sharing one SQLite database.

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## The standout feature

Live OpenWeatherMap forecasts are classified into typed alert events and automatically adjust asset priority scores — the system stays useful beyond a demo.

# Thank You

GridHealth AI — turning sensor noise into a prioritised, weather-aware maintenance plan.

## TEAM TELEMETRY TITANS

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